2 Classical statistical physics – examples

The problems are roughly in order of difficulty. Those indicated with ♡ are ones which are immediately useful practice. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

♡*Problem 2.1* **Average potential energy for a harmonic oscillator**
A particle of mass m resides in one dimension, experiencing a potential $\frac{1}{2}Kq^2$.

(i) Write down the partition function.
(ii) Evaluate the partition function.
(iii) Write down the probability density, $p(q, p)$.
(iv) Evaluate the average potential energy, $\langle V(q)\rangle$.

♡*Problem 2.2* **Energy fluctuations**
We will calculate the size of the fluctuations of the energy of a system in equilibrium with a heat bath at temperature T.

Firstly, using the result for a Gaussian integral show that (by Feynman trick or otherwise)

$$\int_{-\infty}^{\infty} \mathrm{d}x\ x^2 \mathrm{e}^{-\alpha x^2} = \frac{1}{2}\sqrt{\frac{\pi}{\alpha^3}}\ .$$

Use this result, and the approximate result Eqn. (4.4), to show that

$$\langle (E-U)^2\rangle = k_{\mathrm{B}} T^2 C_V\ ,$$

and so deduce that $\sqrt{\langle (E-U)^2\rangle} \propto \sqrt{N} \ll U \propto N$.

♡*Problem 2.3* **Pressure and internal energy for ideal gas**
In lectures we derived the free energy of the ideal gas:

$$F(T,V,N) = -Nk_{\mathrm{B}}T\left\{\ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) + 1\right\}\ .$$

where

$$v = \frac{V}{N} \quad \text{and} \quad \lambda_{\mathrm{T}} = \left(\frac{\hbar^2}{2\pi m k_{\mathrm{B}} T}\right)^{1/2}\ .$$

(i) If we take the general canonical partition function:

$$Z(T,V,N) = \int \mathrm{d}E\ W(E)\mathrm{e}^{-\beta E}\ ,$$

with, as usual, $F = -k_{\mathrm{B}}T\ln Z$, then show

$$U = \langle E\rangle = \frac{\partial(\beta F)}{\partial \beta}\ .$$

This complements the derivation thermodynamically at the beginning of section 4.

(ii) Use the expression in last item to derive the internal energy, U of the ideal gas and hence the specific heat at constant volume, C_V.

♡*Problem 2.4* **Spin-1/2 paramagnet**

In the lectures we derive the (magnetic Gibbs) free energy, $G_{\text{M}}(T, H, N)$, of the spin-1/2 paramagnet, with magnetic moment (including a factor of μ_0), μ in an external field H:

$$G_{\text{M}}(T, H, N) = -k_{\text{B}}T \ln Z = -Nk_{\text{B}}T \ln\left[2\cosh(\beta\mu H)\right] \ .$$

Derive the entropy of the spins, S.

Problem 2.5 **Spin-1 paramagnet**
Let there be N spin-1 atoms, with their spin state denoted $\sigma_n = -1, 0, 1$ for $1 \leq n \leq N$. It has an associated magnetic moment, μ (which incorporates a factor of μ_0), and has interaction with an external magnetic field, H:

$$\mathcal{H} = -\mu H \sum_{n=1}^{N} \sigma_n \ .$$

(i) Write down the variable expressing the magnetisation for this problem.

(ii) Write down the canonical partition function, $Z(T, H, N)$ for the N spins.

(iii) Calculate the average magnetisation, m and isothermal susceptibility, χ_T for the set of spins-1.

Problem 2.6 **Elastic band**

Consider N rods (which we assume are massless and can move through each other) of unit length joined together, so that if the assembly is stretched out it has a length N. See Fig. (2.1).

One end of the set is attached to a support and the other hand has a mass M attached.

Note that the distance between the two ends of the chain is L and this is related to the number of up, $n_\uparrow$, and down, $n_\downarrow$, rods:

$$L = (n_\uparrow - n_\downarrow) \ .$$

(i) Write down the number of configurations with L, $W(N, L)$, and equivalently $W(N, n_\uparrow)$ and $W(n_\uparrow, \downarrow)$.

(ii) What is the potential energy of the weight as a function of L?

(iii) Write down the canonical partition function. Evaluate it noting the analogy with the spin system in example 2, section 5.4 of the lecture notes. (Remember the binomial aspect.)

(iv) Determine $\langle L \rangle$ as a function of T. Interpret this in terms of Hooke's Law ie the displacement of a spring under the action of a force, $f = -Kx$. What is K here?

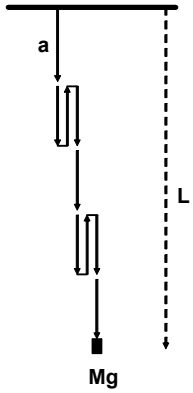

Figure 2.1: Chain of rods hanging under the influence of a weight.

(v) Which parts of the free energy are in balance at equilibrium?

♡*Problem 2.7* **Counterpart to equipartition for different potential shapes**
(A memento from Temperature and matter sheet 5.)
The aim is to derive the counterpart to $\frac{1}{2}k_{\mathrm{B}}T$ per degree of freedom for potential energy of the form $V_n(q) = V_0|q|^n$, with $n > 0$ and $V_0 > 0$.

The corresponding partition functions, Z_n:

$$Z_n = \int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\mathrm{B}}T}\right) \ .$$

And the average value for the potential energy is

$$\langle V_n \rangle = \frac{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}q \ V_0|q|^n \ \exp\left(-\frac{V_0|q|^n}{k_{\mathrm{B}}T}\right) \ \mathrm{d}q}{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\mathrm{B}}T}\right)} \ .$$

We will break the derivation of this quantity into a number of stages.

(i) Firstly a general result on Z_n. Let us define $\beta = 1/(k_{\mathrm{B}}T)$. Then show, using the chain rule, that

$$-\frac{\partial \ln Z_n}{\partial \beta} = \langle V_n \rangle \ . \tag{2.1}$$

(ii) Rewrite the the expression for Z_n, by defining a new variable of integration $y = (\beta V_0)^{1/n}\, q$ to show

$$Z_n = (\beta V_0)^{-1/n} \int_{-\infty}^{\infty} \mathrm{d}y \ \mathrm{e}^{-|y|^n} \ . \tag{2.2}$$

(iii) Now use Eq. (2.1) and apply to Eq. (2.2). Hence show that

$$\langle V_n \rangle = \frac{1}{n} k_{\mathrm{B}}T \ .$$

What value of n corresponds to the harmonic oscillator?

(iv) What is the result for particles in the "flat Earth" gravitational potential, $V(z) = \mathrm{g}z$ for $z \geq 0$, which we considered Y1 Temperature and Matter in analysing the density profile of the atmosphere?

♣*Problem 2.8* **Entropic spring.**
Consider the system whose cross section is shown in Fig. (2.2). This system displays temperature-dependent competition between the internal energy of the (mechanical) spring and the entropy of the hinged rods.

The spring constant of the massless spring is K and it is unstretched when the rods are horizontal. The coupled rods (of length a and mass density ρ per unit length) are freely hinged at each end and at their join. Calculate the partition function of the system, and hence the average spring extension as a function of T.

Where does the entropy reside in this system?

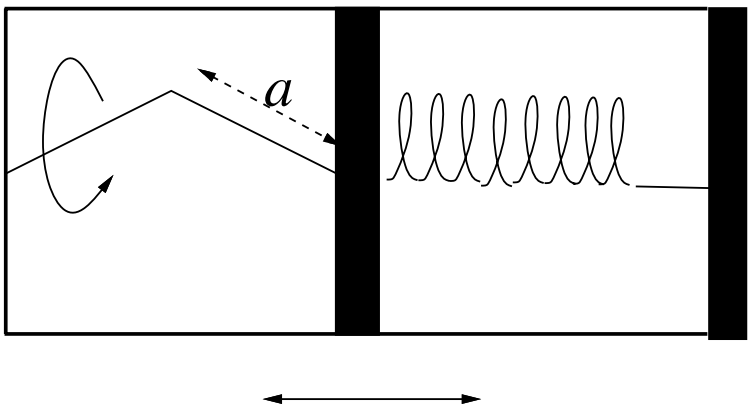

Figure 2.2: The entropic spring.